Nithusikan Thasavaran

AI / ML Engineer

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Profile

AI/ML Engineer building production LLM applications. Currently shipping LLM-powered clinical documentation features (speech-to-text, patient notes, consent letters) for a UK healthcare software company, and built a full-stack RAG system on AWS with measured retrieval quality and latency. Background in end-to-end ML pipelines (Scikit-learn, XGBoost, PySpark, MLflow) and FastAPI/Django backends.

Experience

IN4M Solutions Ltd, United Kingdom — **Aug 2026 – Present**

AI Engineer (Freelance) — *Remote*

- Built a multi-stage **clinical note generation** pipeline that transcribes dental consultations with **gpt-4o-transcribe** and structures them into patient notes with **GPT-4o**; prompt iterations raised clinical checklist coverage from **26% to 60%** (diagnosis section 7% to 64%, risk assessment 29% to 78%) and eliminated template-default leakage and "must-not" clinical errors.
- Re-engineered the **consent letter generation** pipeline (**gpt-oss-120b** via Groq): raised real-case pass rate from **5/8 to 8/8**, achieved **10/10** on every accuracy check across repeated runs, and added automated verification of treatment prices in every letter.
- Cut prompt size by 14% and reasoning tokens by **55%**, and replaced a second LLM "polish" call with deterministic code formatting, bringing every request under the 8K tokens-per-minute limit; replaced generic HTTP 500 failures with automatic retry and clear HTTP 429 responses.
- Benchmarked speech-to-text models (**gpt-4o-transcribe**, **gpt-4o-transcribe-diarize**) on consultation audio; both features are in production after client and clinician sign-off.

Nanthi Ventures Inc. — **Feb 2026 – Apr 2026**

Software Engineer — *Colombo, Sri Lanka*

- Built backend modules for a cloud SaaS platform with **Django**, **PostgreSQL** and REST APIs (authentication, billing workflows, email notifications, query optimization), plus **React**/**TypeScript** frontend components in an agile team.

LIRNEasia — **Jul 2024 – Jan 2025**

Data Science Intern — *Colombo, Sri Lanka*

- Conducted NLP research on ~**100K multilingual tweets** (English, Tamil, Sinhala) to predict tweet virality from textual and metadata features.
- Built an end-to-end ML pipeline (scraping with **Scrapy/Selenium**, preprocessing, feature engineering, tuning, evaluation) tracked with **MLflow** and **DVC**, benchmarking Logistic Regression, Random Forest, XGBoost and TF-IDF/Word2Vec/FastText features.
- Performed hashtag network analysis with **NetworkX**, **Neo4j** and community detection, with interactive visualizations.

Projects

Anamnesis – Production RAG Medical Study Assistant | *Individual* | 2026 Jun – Present

- Deployed a full-stack RAG application on **AWS**: React + TypeScript on **S3 + CloudFront**, containerized FastAPI backend on **ECS Fargate** (ECR), **RDS PostgreSQL**, and **Pinecone**, with **GitHub Actions** CI/CD; serves Gemini Flash and open-weight models (gpt-oss-120B/20B, Qwen).
- Built history-aware hybrid retrieval (dense + BM25) with **cross-encoder reranking** in my own LangChain-based RAG library; end-to-end evaluation with **RAGAS**.
- Ran retrieval evaluation (**Recall@K, MRR, nDCG**, paired-bootstrap CIs) over a 6,665-chunk textbook index with 150 labelled queries: production reached **Recall@5 0.94** and **MRR 0.80**, **+41 points** Recall@5 over BM25; ablations showed reranking adds +12 points and equal-weight fusion underperforms dense-only, guiding the next retrieval change.
- Designed a two-tier response cache: exact hits return in **0.14 ms vs 3.3 s** for the full pipeline and save ~1,340 tokens per repeat; threshold calibration found the semantic tier too strict to fire and identified a safer setting.

Cross-Layer Cyberattack Detection Framework | *Final Year Project* | 2025 Mar – Aug

- Built an ML framework to detect known and unknown cyberattacks by correlating network traffic with application-layer logs across **12M+ flow records**, with **PySpark** preprocessing to handle high dimensionality and class imbalance.
- Trained multiclass detectors with **Random Forest** and **XGBoost** reaching up to **97% accuracy** (tracked in **MLflow**), plus semi-supervised anomaly detection for previously unseen attacks.

Technical Skills

Languages: Python, Java, SQL, JavaScript, TypeScript, C

LLMs & GenAI: LangChain, OpenAI API (GPT-4o, gpt-4o-transcribe), Gemini, gpt-oss, Llama 3, Qwen, Groq, Hugging Face Transformers, Prompt Engineering, Speech-to-Text

RAG & Evaluation: Pinecone, ChromaDB, FAISS, Sentence Transformers, BM25, Cross-Encoder Reranking, RAGAS, Recall@K / MRR / nDCG

Machine Learning: Scikit-learn, PyTorch, TensorFlow, XGBoost, YOLOv8, NLP, Computer Vision

Data & MLOps: Pandas, NumPy, PySpark, MLflow, DVC, Scrapy, Selenium, BeautifulSoup

Backend, Web & Databases: FastAPI, Django, Spring Boot, REST APIs, React, PostgreSQL, MySQL

Cloud & DevOps: AWS (ECS Fargate, ECR, RDS, S3, CloudFront, EC2), Docker, GitHub Actions, Git

Education

University of Peradeniya, Sri Lanka — **2021 March – 2025 August**

BScEng(Hons) specializing in Computer Engineering — *3.50 / 4.00 CGPA*

Certifications & Achievements

- Stanford (Coursera): Supervised Machine Learning, Advanced Learning Algorithms — Java – edX, LinkedIn Learning
- ACES Hackfest 2023 – **Winners**, Sustainable Development Category (AI/ML) — ACES Coders v10.0 – **Finalist**